Prateek Awate

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Research Interests

Primary: Quantum error correction, especially machine-learning decoders that are accurate and fast enough for real-time use.
Related: Decoding as statistical inference, data-efficient learning, quantum optimization.

Education

Degree	Institution	Dates
MS, Computer Science (online, OMSCS)	Georgia Institute of Technology, Atlanta, USA	Expected Dec 2027
B.E., Electronics and Telecommunication	Dr. D. Y. Patil School of Engineering, Pune, India	May 2021

Publications

1. **Prateek Awate.** Cluster-Aware QUBO Instance Selection with Representativeness Scoring and Noise Pre-Filtering for the QuantumCLEF 2026 Sentiment Classification Task. *CLEF 2026 Working Notes* (CEUR-WS), Jena, Germany, Sept 2026. Solo author. 1st among pure quantum annealing submissions, 4th overall in Task 2.
2. R. Taylor, A. Brikman, **Prateek Awate.** Searching for Sound-Meaning Collisions: Graph-Based Affordance Retrieval and Multi-Evaluator Ranking for Cross-Lingual Pun Translation at CLEF 2026 JOKER Task 2. *CLEF 2026 Working Notes* (CEUR-WS), Jena, Germany, Sept 2026.
3. **Prateek Awate,** Avinash Yadav, Sai Kumar, Nishikant Surwade, Rashmi Mahajan**.** Machine Learning Applications for Predicting Cardiac Events. *International Research Journal of Modernization in Engineering Technology and Science (IRJMETS)*, 2021.

Research Experience

QUBO instance selection on quantum annealing hardware - QuantumCLEF 2026, Task 2 (DS@GT, Georgia Tech) · 2026

- Formulated training-instance selection for noisy sentiment data as a QUBO with a cluster-aware pairwise penalty and a representativeness-aware diagonal (SVM margin plus class-centroid proximity).
- Removed likely mislabelled instances before optimization with a Cleanlab confident-learning pre-filter; tuned hyperparameters with multi-objective Bayesian optimization (BoTorch).
- Ran the same QUBO with simulated annealing and on D-Wave Advantage. The QA run reached Macro-F1 64.8 ± 8.9 at 67.6% data reduction: 1st among pure quantum annealing entries, 4th overall, and the only team where quantum annealing beat simulated annealing (by 4.4 points).

Retrieval-augmented pun translation - CLEF 2026 JOKER, Task 2 (team of three) · 2026

- Team system combining semantic retrieval (bge-m3, FAISS), learned phonetic embeddings and two-stage multi-model generate-and-rank with weighted Borda voting; first on the public leaderboard at the time of writing.
- My contribution: helped design the core pun-translation method.

LITE Batching: lightweight training-batch selection – Georgia Tech CS6999 · Spring 2026 · supervised by Prof. Steve Mussmann

- Benchmarked 8 training-batch selection methods for reducing wasted GPU compute, across multiple datasets and settings.
- Project ended with the semester before a conclusive result.

Professional Experience

Senior AI Engineer - Pridenexus Pvt Ltd, Pune · Dec 2025 - Aug 2026

- Prototyped a vision-language document pipeline: fine-tuned YOLOv8n for region detection, MobileSAM segmentation and a LoRA-fine-tuned SmolVLM; evaluated zero-shot vs fine-tuned models with precision, recall, F1 and mAP.
- Designed LLM services (FastAPI, PostgreSQL, Weaviate retrieval) and mentored 3 engineers.

AI Engineer - Technocraft Solutions, Mumbai · Dec 2024 - Dec 2025

- Built a real-time, fully on-device identification system on Raspberry Pi: YOLO person detection plus InsightFace recognition (99%+ precision).
- Deployed an air-gapped retrieval-augmented LLM (Llama 3 8B, BF16) on an NVIDIA RTX 4090, with source attribution over confidential documents.

Associate Software Engineer - Mphasis Ltd, Mumbai · Dec 2021 - Feb 2024

- Built data pipelines (SQL, Pandas, NumPy) and tuned database performance for enterprise clients.

Assistant Tech Support - Shri Girijatmak Pvt Ltd, Pune · Apr 2021 - Dec 2021

Technical Skills

- **Programming:** Python (NumPy, Pandas, async), SQL, TypeScript
- **Machine learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LoRA fine-tuning, retrieval (FAISS, Weaviate)
- **Optimization and quantum:** QUBO formulation, D-Wave Ocean SDK (quantum and simulated annealing), Bayesian optimization (BoTorch); learning Stim and PyMatching
- **Data and evaluation:** confident learning (Cleanlab), cross-validation, precision/recall/F1/mAP
- **Edge and systems:** on-device inference (Raspberry Pi, NVIDIA Jetson, RTX 4090), Docker, Git

Certifications

- **DP-900** Microsoft Certified: Azure Data Fundamentals
- **NVIDIA** - Building LLM Applications with Prompt Engineering
- **NVIDIA** - Rapid Application Development using Large Language Models

Languages

English (Fluent) · Hindi (Native) · Marathi (Native)